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THE USE OF LINEAR POTENTIAL SWEEP VOLTAMMETRY AND OF A.C. VOLTAMMETRY FOR THE STUDY OF THE SURFACE ELECTROCHEMICAL REACTION OF STRONGLY ADSORBED SYSTEMS AND OF REDOX MODIFIED ELECTRODES

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ABSTRACT

The use of linear potential sweep (l.p.s.) voltammetry and of a.c. voltammetry for the study of the surface electrochemical reaction of strongly adsorbed systems and of redox modified electrodes is discussed. Ideal l.p.s. or a.c. voltammograms are obtained when there are no interactions between the molecules, the electrode surface is homogeneous and the electrochemical reaction is at equilibrium (v/k_s or $f/k_s \rightarrow 0$; v = sweep rate; f = frequency; k_s = rate of the electrochemical reaction). The l.p.s. peaks are symmetrical and the oxidation and reduction peaks have the same peak potential; in a.c. voltammetry, cotg ϕ (ϕ = phase angle) is equal to $\pi/2$. Interactions between the molecules result in a shift of E_p and in a change of the shape of the curves; in l.p.s. voltammetry the oxidation and reduction peak still have the same peak potentials, and in a.c. voltammetry cotg ϕ is still equal to $\pi/2$. Heterogeneity of the electrode surface should have effects similar to those observed under certain conditions for interactions between the molecules. When v/k_s or f/k_s increases, a difference between the l.p.s. oxidation and reduction peaks appears, and in a.c. voltammetry cotg ϕ increases, which allows k_s to be calculated. Particular applications to three types of electrodes, modified by irreversible adsorption, by chemical binding and by coating by a polymer are discussed.

INTRODUCTION

There has been recently a growing interest in obtaining and in studying modified electrodes, which are prepared by attaching irreversibly chemical species to the surface of a conducting material [1—21,34]. The species can be electroinactive [1—4,18], or can consist of a redox couple [5—17,19,20,34] with the surface redox reaction $0 + ne \rightleftharpoons R$ taking place (redox modified electrodes). Three principal methods have been used for preparing modified electrodes: irreversible adsorption [5—10], coating (especially by polymers) [4,19—21], and chemical binding [1—3,10—18,34]. Among the possible applications of these electrodes, asymmetric syntheses [1,18] and electrocatalysis [8,9,12,13] have been more particularly considered.

A knowledge of the characteristics of the surface electrochemical reaction in the case of strongly adsorbed systems and of redox modified electrodes is interesting for theoretical reasons. It is also desirable for the study of the

electrocatalytic properties of modified electrodes; let us consider for example the general catalytic scheme:

$$O + ne \rightleftharpoons R \qquad (I)$$

$$A + R \rightarrow O + B \qquad (II)$$

If the rate of reaction (I) is slow enough, it can appear as a controlling factor in the total rate of the catalytic process.

Linear potential sweep voltammetry (l.p.s. voltammetry) has been widely used for the qualitative examination of redox modified electrodes. We will discuss here its use for the quantitative study of the equilibrium and kinetic properties of attached redox systems, and we will show that a.c. voltammetry (or faradaic impedance measurements) can be used advantageously to the same purpose. In what follows double layer effects will be neglected. We will first consider systems with strong adsorption of O and R, and then the application of the results to the study of modified electrodes.

CASE OF A STRONG ADSORPTION OF O AND R

(1) Equilibrium of the surface reaction

Both the adsorption reaction and the electrochemical reaction are assumed to be at equilibrium, i.e. the rate of these reactions does not control the current [22,28].

(A) Ideal system. By definition, we will consider as ideal the case for which: (a) all the adsorption sites are equivalent; (b) there are no interactions between the adsorbed molecules. These conditions correspond e.g. to a Langmuir isotherm. They can be fulfilled experimentally by using a mercury electrode (the homogeneousness of the surface ensures that the adsorption sites are equivalent), when the superficial concentrations tend toward zero (the interactions which eventually exist between the molecules become negligible).

As we have shown in ref. 22, a surface standard potential $E^{0\prime}$ can be defined, whose expression is:

$$E^{0\prime} = E^0 - (RT/nF) \ln(b_O/b_R) \qquad (1)$$

in which E^0 is the standard potential of the species in solution, and b_O and b_R the adsorption coefficients of O and R. b_O and b_R are assumed to be independent of the potential, which is justified as a first approximation if the range of potential examined is not too large, e.g. of the order of a tenth of a volt [28].

The superficial concentrations Γ_O and Γ_R of O and R obey a surface Nernst equation:

$$\Gamma_O/\Gamma_R = \exp[(nF/RT)(E - E^{0\prime})] = \eta \qquad (2)$$

The meaning of eqn. (1) is clear. We can write [23]:

$$b_O = \exp(-\Delta G_O/RT) \qquad (3)$$

$$b_R = \exp(-\Delta G_R/RT) \qquad (4)$$

in which ΔG_O and ΔG_R are the standard free energies of adsorption of O and R. Equation (1) can be written:

$$nF(E^{0\prime} - E^0) = \Delta G_O - \Delta G_R \tag{5}$$

which shows that the increase or decrease of electrical energy which has to be spent during the reduction or oxidation is equal to the difference of the free energy of adsorption of O and R. If, O for example, is strongly adsorbed, ΔG_O decreases, so that $E^{\prime 0}$ becomes more negative than E^0; in the case of a reduction, more electrical energy has to be spent in order to compensate for the energy lost by the molecule during the adsorption process.

We have established in previous papers [22] the expression of the l.p.s. voltammogram, and in refs. 24 that of the a.c. peak (or faradaic impedance) in the case considered here when the adsorption obeys a Langmuir isotherm. The results can be expressed by using the dimensionless function:

$$|\Psi_v| = |i|/(F^2n^2vA\Gamma_T/RT)$$

in the case of l.p.s. voltammetry, and

$$\Psi_{ac} = i/(F^2n^2\omega\epsilon A\Gamma_T/RT)$$

for the a.c. peak, where i is the current, A the electrode area; Γ_T the total superficial concentration, v the sweep rate, ϵ the amplitude of the alternating potential, and ω the angular frequency.

The equation is the same in both cases:

$$|\Psi_v| = \Psi_{ac} = \eta/(1 + \eta)^2 \tag{6}$$

The peak potential E_p is equal to $E^{0\prime}$; the peak has a symmetrical shape, and its width at mid-height δ is equal to $90.6/n$ at 25° C (Fig. 1, $G = 0$, or Fig. 3, $nvF/RTk_s = 0$). In the case of l.p.s. voltammetry, the oxidation and reduction peaks are symmetrical (their peak potential is the same) (Figs. 1 and 3). In a.c. voltammetry, the phase angle is equal to $\pi/2$.

(B) Non-ideality caused by interactions between the adsorbed molecules. This can be the case for example with a mercury electrode, when the superficial concentrations are large. The interactions between the molecules can be taken into account by using for example a Frumkin type isotherm [25—27].

As we have shown in previous papers [28], eqn. (2) is no longer valid, and must be replaced by another expression. If we assume that a Frumkin isotherm is obeyed and that the area occupied by one molecule of O is equal to the area occupied by one molecule of R, this expression is:

$$\eta = (\theta_O/\theta_R)\exp[2v\theta_O(a_{OR} - a_O) + 2v\theta_R(a_R - a_{OR})] \tag{7}$$

In this equation and in the following, a_O, a_R and a_{OR} are the constants of interaction between molecules of O, molecules of R and molecules of O and R, respectively (a_i is positive for an attraction and negative for a repulsion; the a values are assumed to be independent of the potential [28]. v is the number of molecules of solvent (or clusters of solvent molecules [27,29]) displaced by one molecule of O or R. θ_O, θ_R and θ_T are respectively the coverages of O, R and

O + R ($\theta_O = \Gamma_O/\Gamma_m$, $\theta_R = \Gamma_R/\Gamma_m$, $\theta_T = \Gamma_T/\Gamma_m$, Γ_m being the maximal value of either O or R).

In ref. 28, we have presented the theory of l.p.s. voltammetry under these conditions. The peak potential is given by:

$$E_p = E^{0\prime} - (RT/nF)(a_O - a_R)\nu\theta_T \quad (8)$$

The meaning of this equation is clear: $RTa_O\nu\theta_T$ represents one half of the energy which has to be spent, or which is released during the total reduction or oxidation because of the gradual disappearance or appearance of the interaction forces between molecules of O (at the peak potential, one half of the molecules of O have appeared or disappeared).

If for example, $a_O > 0$ (attraction forces between molecules of O), the peak potential becomes more negative; in the case of a reduction, for example, more electrical energy has to be spent in order to overcome the attraction forces. The term in $RT\nu a_R\theta_T$ can be interpreted similarly. A more detailed discussion will be given in a subsequent paper [30].

The shape of the peak is defined by the quantity $\nu G\theta_T = \nu\theta_T(a_O + a_R - 2a_{OR})$ (Fig. 1).

If $\nu G\theta_T = 0$, the peak has the shape and the characteristics of the ideal peak.

If $\nu G\theta_T < 0$, the peak becomes broader and flatter.

If $0 < \nu G\theta_T < 2$, the peak becomes narrower and sharper.

In the three cases, the peaks still have a symmetrical shape, and the oxidation and reduction peaks are still symmetrical, with the same peak potential (Fig. 1).

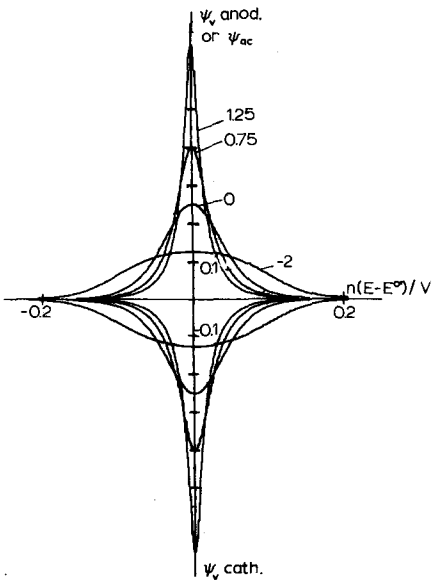


Fig. 1. Equilibrium l.p.s. voltammetric peaks and a.c. peaks at 25°C in the case of a Frumkin isotherm when $\nu G\theta_T < 2$. The value of $\nu G\theta_T$ is shown on each curve.

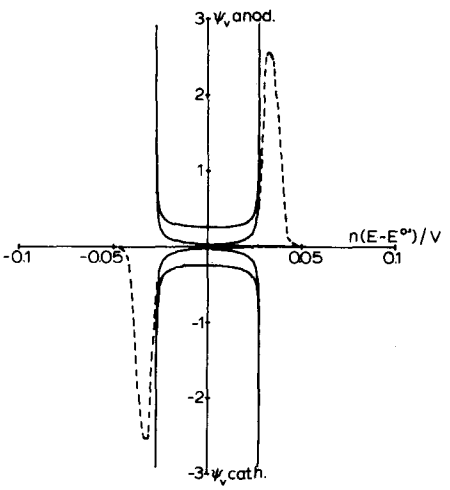


Fig. 2. Equilibrium l.p.s. voltammetric peaks and a.c. peaks at 25°C in the case of a Frumkin isotherm when $\nu G\theta_T > 2$ ($\nu G\theta_T = 4$). (———) Theoretical curve; (- - - - -) actual peaks (they have been given an arbitrary shape; see text).

If $\nu G\theta_T > 2$, the theoretical expression of the peak consists of three branches (Fig. 2), which can in certain cases be obtained experimentally, at least partly [30]; usually, however, during a cyclic potential scan, a hysteresis phenomenon appears: the peaks no longer have the same peak potential (Fig. 2). The shape of the peaks can no longer be described mathematically.

It must be pointed out that if an ideal peak is obtained, this does not necessarily mean that no interaction forces exist, since $G = 0$ can result from a compensation between the interaction constants.

In the case of a.c. voltammetry, the theory [30] shows that the peak has the same shape as the corresponding l.p.s. voltammogram, (Fig. 2) and that the phase angle is still equal to $\pi/2$.

(C) Non-ideality caused by the inhomogeneity of the electrode. In the case of solid electrodes, the non-homogeneous character of the surface has to be taken into account. The effect on the adsorption equilibrium can be described by using for example a Temkin isotherm [27a,31]. When the superficial concentration tends towards zero, this isotherm reduces to the Henry isotherm [31]. The theory for l.p.s. voltammetry and for a.c. voltammetry has not yet been developed. Some qualitative predictions can however be made.

Temkin made the hypothesis that the adsorption of the first molecules takes place on certain sites (favourable sites), where the interactions between the molecules and the surface are the greatest, i.e. where the energy of the molecule is the lowest. Then less and less favourable sites, i.e. sites where the energy of the molecule increases, are occupied. In the derivation of the isotherm, Temkin assumed that the energy of the molecules increases linearly with the coverage.

We will consider a reduction (the same reasoning can be made for an oxidation) and we will distinguish two cases. Let us first assume that the reduction of a molecule of O will give a molecule of R on the same site; this situation will hold for example at low coverage, when the distance between the adsorbed molecules is large. If the coefficients of the Temkin isotherm relative to O and R are equal, i.e. if the effects of the heterogeneity is the same on O and R, the voltammetric peak will have the ideal shape, since the difference in energy between O and R is constant. If, on the contrary, the coefficients are different, the reduction will first take place on the sites for which the difference of energy between O and R is the lowest, then it will gradually occur on sites where the difference of energy is higher and higher: more and more electrical energy has to be used when compared to the ideal case, i.e. the peak will be wider and flatter.

If it is assumed that electron exchanges are possible between the adsorbed molecules, i.e. for example when the coverage is high, so that the molecules are close to one another, another scheme is possible. Reduction of molecules of O on "unfavourable" sites will give molecules of R on "favourable" sites, since the difference in energies is then the lowest. Then molecules of O with a lower energy (on more "favourable" sites) will give molecules of R with a higher energy (on less "favourable" sites); here again more electrical energy than in the ideal case has to be used, so that the peak will also be broader and flatter.

It can thus be concluded that the heterogeneities of the surface will always cause the peak to be broader and flatter than the ideal peak. If a Temkin iso-

therm is assumed to hold, the shape of the peak should not be very different from that of the peak obtained with a Frumkin isotherm, when $G < 0$: the shape of the Temkin isotherm is practically indistinguishable from the shape of the Frumkin isotherm in which an adequate negative value of the interaction coefficient is introduced, although their mathematical expressions are different. Besides, in the derivation of both isotherms it is assumed that the energy of the molecules varies linearly with the coverage.

When interactions between the molecules are present, the situation becomes more complicated (an isotherm taking into account the effects of the inhomogeneities of the electrode and the interaction forces has been derived by Batrakov and Damaskin [32]). The effects noted above on the width of the peak should be cumulative. In particular the possibility cannot be excluded that a peak having an ideal or nearly ideal shape may be the result of a compensation between the effects of the inhomogeneities of the electrode and of the existence of a positive G parameter.

(2) Kinetics of the surface reaction

(A) No interactions between the molecules and homogeneous surface. If there are no interactions between the molecules, the general current-potential characteristic can be written (anodic currents are considered as positive)

$$i = -nFAk_s(\Gamma_O\eta^{-\alpha} - \Gamma_R\eta^{1-\alpha}) \tag{9}$$

k_s being the rate of the surface reaction in s^{-1}, and α the transfer coefficient.

At the equilibrium, when $i = 0$, eqn. (2) is obtained, i.e. eqn. (9) is compatible with a Langmuir isotherm.

We have developed in ref. 33 the theory of l.p.s. voltammetry for this case. When $v/k_s \to 0$, symmetrical ideal anodic and cathodic peaks are found (Fig. 3). When v/k_s increases, the shapes of the peaks change, and the difference ΔE_p between the cathodic and anodic peak potential increases. The rate constant k_s can be calculated from the experimental values of ΔE_p and v.

In ref. 24, we have established the theory in the case of a.c. voltammetry. The cotangent of the phase angle, cotg ϕ, is a linear function of the frequency. k_s can be deduced from the slope of the straight line obtained (k_s is inversely proportional to the slope).

(B) Interactions between the molecules and homogeneous surface. The hypothesis can be made that the activation energies of the cathodic and anodic processes are proportional to the coverages θ_O and θ_R. The areas occupied by one molecule of O and one molecule of R are still supposed to be equal. If we assume that k_s is independent of the coverage, the i—E characteristic can then be written [30]:

$$i = -nFAk_s\Gamma_m[\theta_O\eta^{-\alpha}\exp(-\beta\theta_O - \gamma\theta_R) - \theta_R\eta^{1-\alpha}\exp(-\lambda\theta_R - \mu\theta_O)] \tag{10}$$

At equilibrium, when $i = 0$, an expression having the same form as eqn. (7) is found. Equation (10) is thus compatible with a Frumkin isotherm.

We have solved the problem under these conditions for the case of a.c. voltammetry [30]. Cotg ϕ still varies proportionally to the frequency, but the slope of the straight line obtained is a complicated function of Γ_O and Γ_R and of the interaction coefficients β, γ, λ and μ. It does not appear possible for the moment to determine these coefficients [30], so that the determination of the actual

value of k_s will not be possible. Only an apparent rate constant can be deduced from the results; its value can be very different from the actual value of k_s [30].

It is actually possible that the rate constant k_s is not independent of the coverage: it is well known, for example, that when the coverage increases, the orientation of the adsorbed molecules may vary, which could cause a change in k_s. As shown by the foregoing discussion, however, it seems a priori difficult to study eventual changes in k_s, since the interaction coefficients are not known.

(C) Non-homogeneous surface. If the heterogeneity of the surface has an effect on the adsorption, i.e. if the energy of adsorption of the molecules changes with the coverage, changes in the activation energies of the redox reaction are a priori possible. It could be assumed for example that the activation energies are linearly related to the coverage; the consequences would then be more or less similar to those discussed in the case of interactions between the molecules.

APPLICATION TO MODIFIED ELECTRODES

When the surface redox system is attached to the electrode by irreversible adsorption, the application of the results given above is immediate, since they have been derived by assuming [22,28] that the concentrations in solution tend towards zero. As results from the preceding discussion, the value of k_s can be obtained by measurements at several coverages, followed by extrapolation to zero coverage, since this ensures that the influence of the interactions between the molecules and of the non-homogeneity of the electrode becomes negligible (on the condition that the molecules do not form aggregates).

With chemically bonded redox systems, the equations given above can still be assumed to apply. The energies in eqns. (3) to (5) will now represent the energy of the chemical bonds with the electrode. These energies will probably not be very different for the oxidized and for the reduced forms, since the bond between the molecule and the electrode is generally not involved in the redox process. The surface redox potential should not therefore in general be very different from the redox potential in solution. This has indeed been

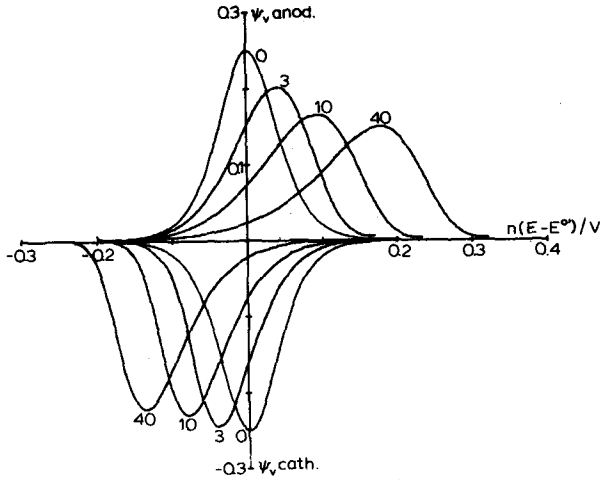


Fig. 3. L.p.s. voltammetric peaks with control by the rate of the electrochemical reaction at 25°C. $\alpha = 0.6$. The value of $n\nu F/RTk_s$ is shown on each curve.

shown to be the case in all the experimental comparisons made until now [9, [9,10,12,14—16,34].

The polymer coated electrodes will pose some special problems. The effect of the heterogeneity of the electrode cannot be excluded a priori. On the other hand there should exist in general strong interactions between the electroactive centers. L.p.s. voltammograms whose shape indicates such interactions have been found experimentally [19—21]. It seems a priori difficult to eliminate these effects, since they are inherent to the structure of the molecule, and since it is not possible to reduce the superficial concentration of the electroactive centers, as in the other two cases. Another problem is that of the possibility of existence of multilayers, since the transmission of the electrons through the layers could obey other rules. Under these conditions, only an apparent rate constant will a priori usually be obtainable.

It has been found in several cases, especially with polymer coated electrodes [19—21] (see however also ref. 34) that the cathodic and anodic l.p.s. voltammetric peaks have a different width, although their peak potential and their area is the same. As shown above, symmetrical peaks should be obtained if the process is reversible. Non-equilibrium phenomena, like slow changes in the structure of the film should then occur during the reduction or oxidation.

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